

The dynamical prior Φ

9,420 parameters · 0.5% of the model

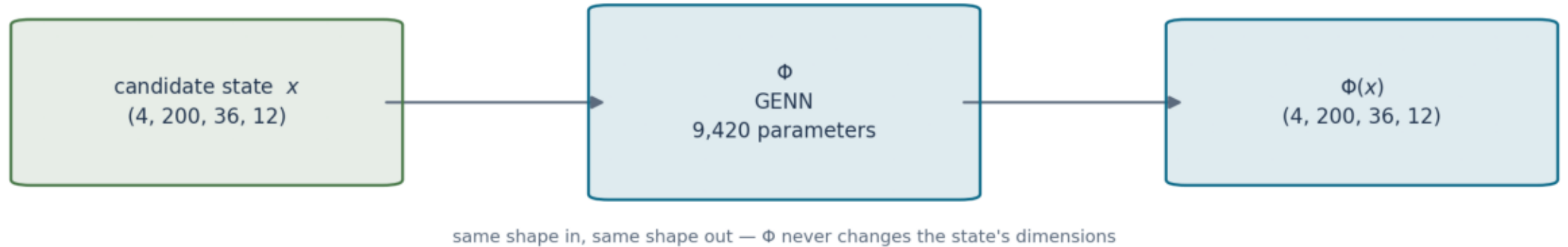
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What the prior is for

$$J(x) = \alpha_{obs}^2 \|(x - y) \odot \Omega\|^2 + \alpha_{reg}^2 \|x - \Phi(x)\|^2$$

fit what the robots measured

stay a plausible crowd field

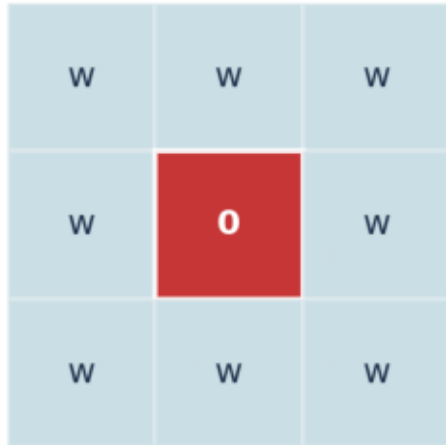


- Not a forecaster: given x , it says what each frame SHOULD look like, judged from the frames around it.
- Plausible $x \rightarrow x \approx \Phi(x)$, term small. Impossible $x \rightarrow$ term large, solver pushed away.
- The only route to cells no robot saw — the observation term is zero there.

Inside Φ , part 1 — ψ , the one convolution that sees neighbours

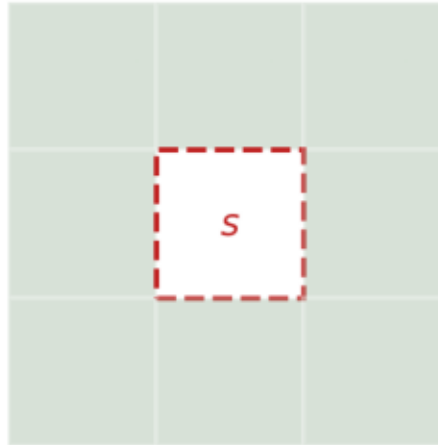
ψ : Conv3d(4 $\rightarrow$ 32, kernel $3 \times 3 \times 3$), 3,488 parameters

centre time slice of the $3 \times 3 \times 3$ kernel



the centre tap is masked to 0
on every forward pass

input read at each neighbour of s



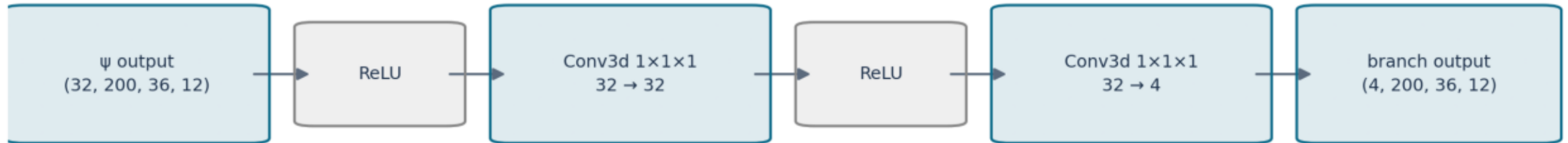
$x(s)$ itself is never read



- The output at a point never reads the input at that point.
- Without it, $\Phi = \text{identity}$ drives $\|x - \Phi(x)\|^2$ to zero for EVERY x — the prior would constrain nothing.
- The mask is re-applied every forward pass, so the optimiser cannot undo it.

Inside Φ , part 2 — φ , pointwise on purpose

φ : 2 pointwise Conv3d layers, every kernel $1 \times 1 \times 1$, 1,188 parameters

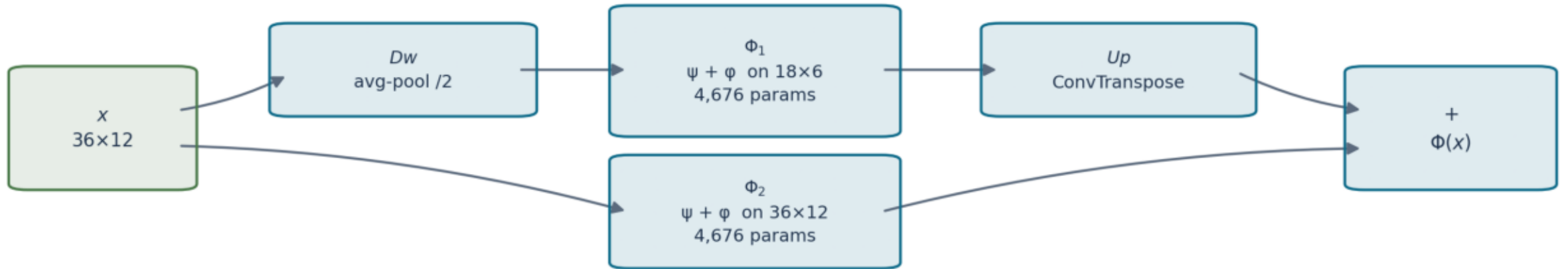


kernel 1 everywhere → each of the $200 \times 36 \times 12 = 86,400$ space-time positions is transformed on its own, using only its own 32 feature values. No neighbouring cell, no neighbouring frame.

- Paper Sec. 3.2 requires it: every kernel size is 1.
- A wider kernel would pull in a neighbour — which does depend on $x(s)$ — and make the identity reachable again.
- So all spatial and temporal context comes from ψ alone. Widening φ adds capacity, not reach.

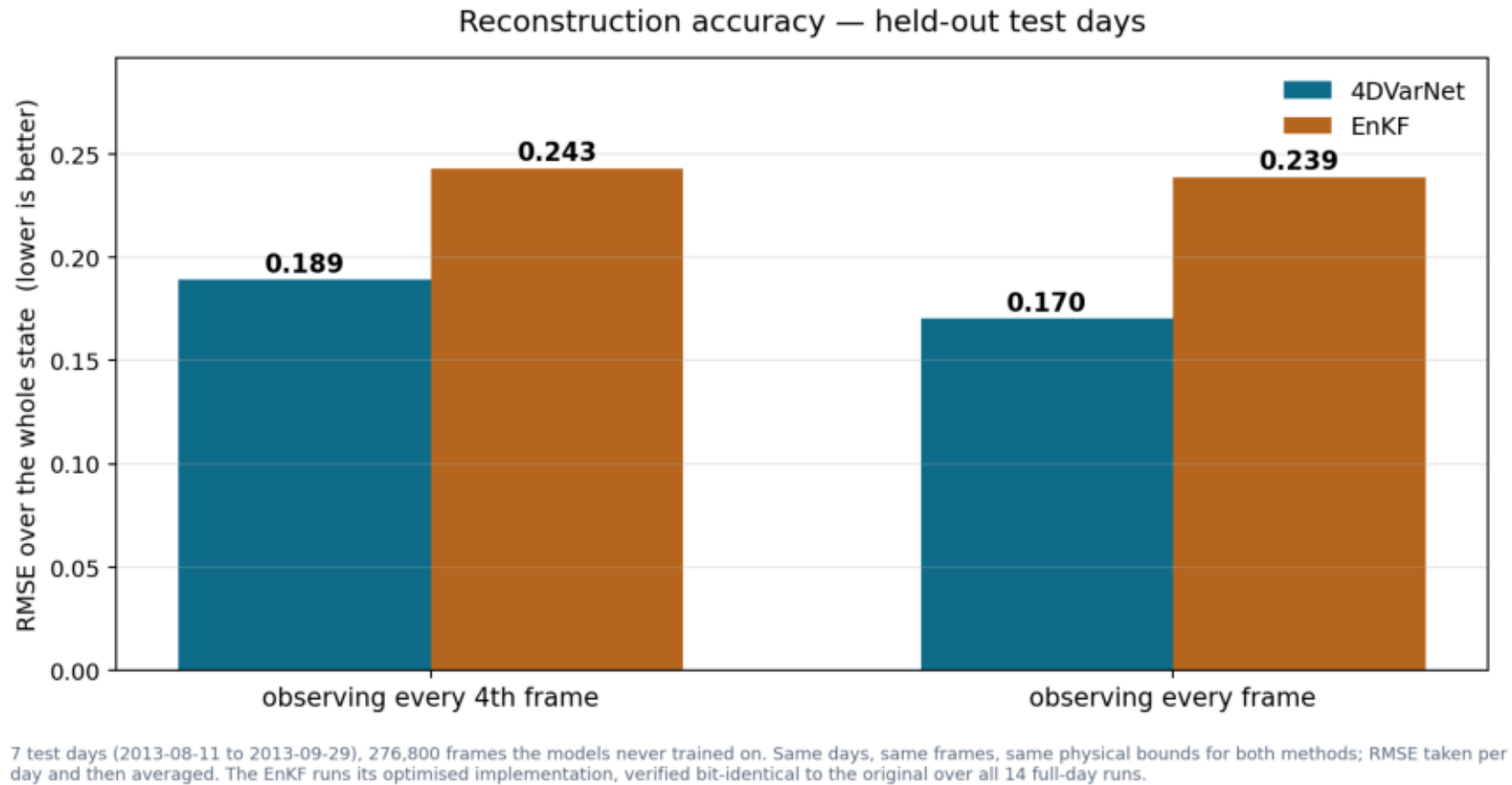
Inside Φ , part 3 — two scales (Eq. 10)

$$\Phi(x) = \text{Up}(\Phi_1(\text{Dw}(x))) + \Phi_2(x)$$



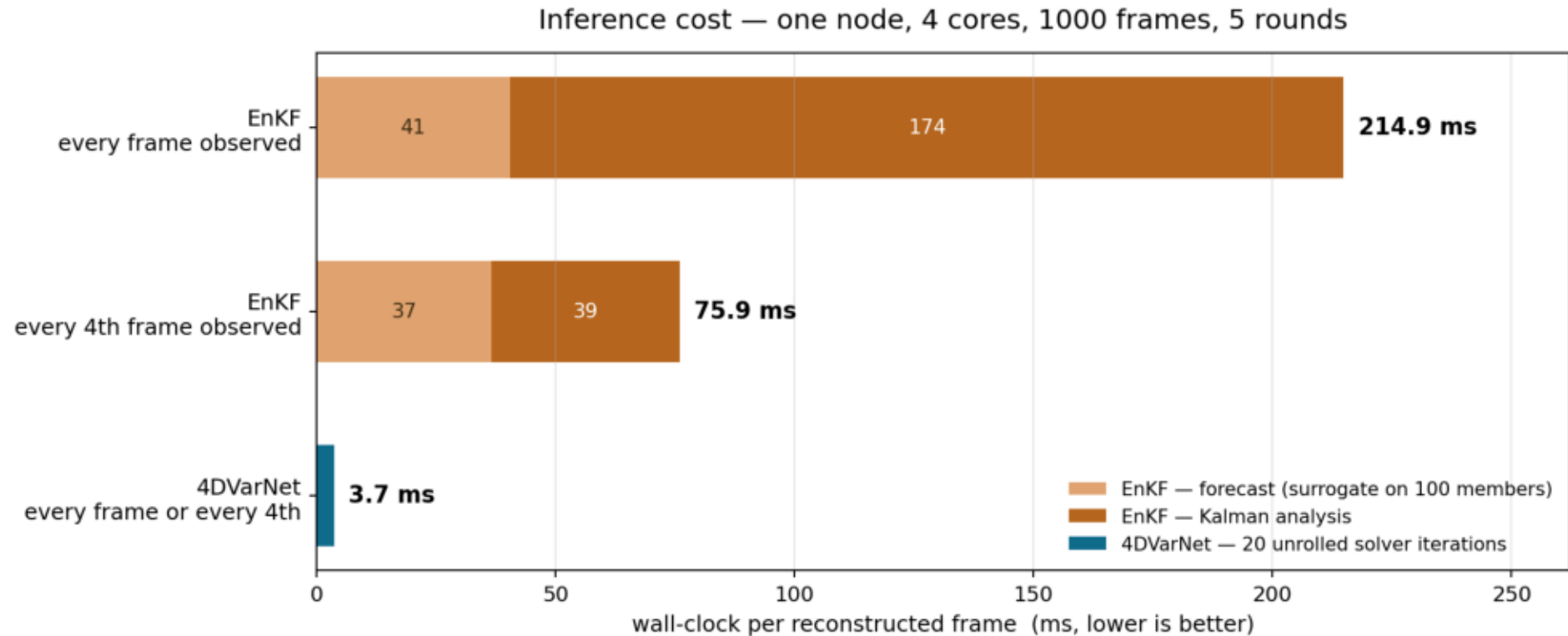
- Each branch is a full $\psi + \phi$ (4,676 params), both zero-centre. Prior total 9,420.
- Φ_1 runs AT 18×6 , so its 3×3 kernel covers 6×6 fine cells — genuinely different scales.
- Φ_2 takes x itself, not a high-pass residual; Up is learned, not interpolation.

Where it stands — accuracy



- 7 held-out test days, 276,800 frames. Same frames and bounds for both methods; RMSE over the whole state.

Where it stands — inference cost



Intel(R) Xeon(R) Gold 6248 CPU @ 2.50GHz · 4 cores, node csl19. Both methods on the same CPU, in one job, interleaved: no GPU is involved, and the EnKF is its optimised implementation. Only the analysis step scales with observation density — the forecast runs on every frame either way — which is why the two EnKF bars differ while 4DVarNet's cost does not (3.66 vs 3.63 ms). Split measured separately under the same conditions. Largest run-to-run spread across the 5 rounds: 18% of the mean — the node was shared, so read these as ratios between methods rather than as absolute constants.

- One node, 4 cores, no GPU either side. EnKF is its optimised implementation, bit-identical to the original.